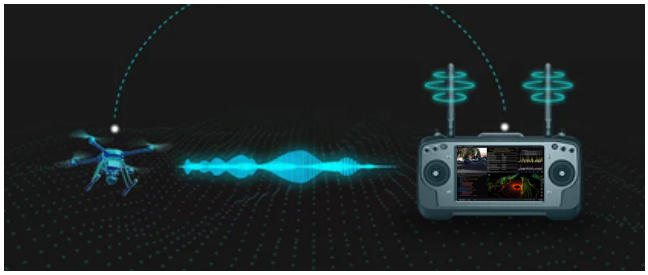
TRAx Controller

Terrain-Ready Autonomy Extension Controller

OVERVIEW

Easily transforms conventional vehicles into autonomous or remotely controlled systems, enabling teleoperation and seamless integration of various sensors for advanced perception and control.

The TRAx Controller requires minimal vehicle modifications and supports real-time actuator control, making it ideal for retrofitting ground vehicles for research, testing, or field deployment in challenging environments.



FEATURES

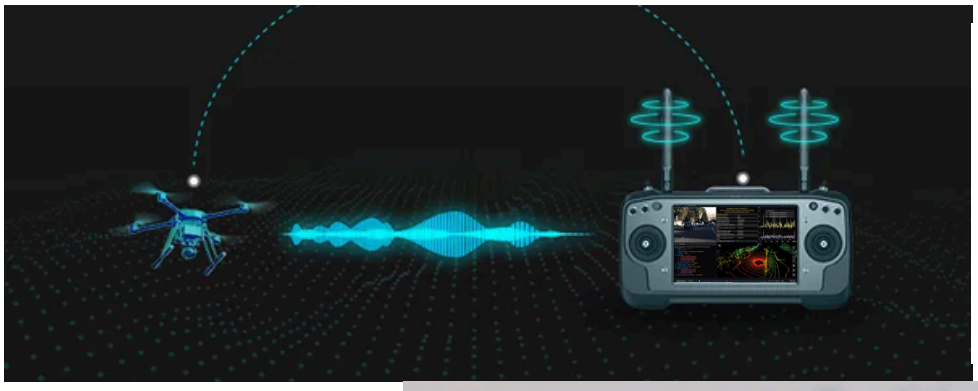
Vehicle Conversion	Converts regular IC engine-based vehicles into autonomous vehicles.
Dual Mode Control	Supports both autonomous stack control and RC (remote control) operation.
Closed-loop Feedback System	Position feedback system ensures precise actuator control. Enables closed-loop control for accurate navigation.
Sensor Compatibility	Easily connect custom sensors (e.g., LIDAR, camera, IMU) for perception and localization tasks.
Advanced Velocity Control	Employs a neural network-based PID controller to fine-tune velocity and maintain constant speed.

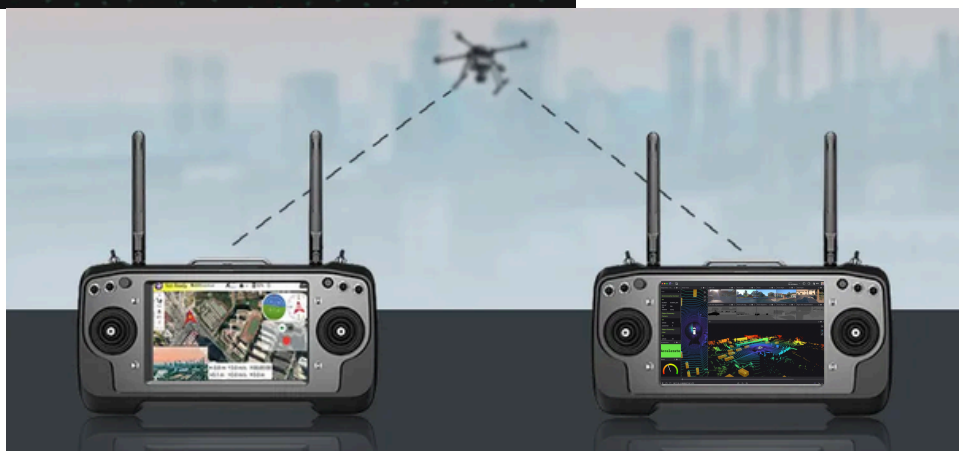
APPLICATIONS

- **Remote Vehicle Operation**
- **Night time operation**
- **AI Based Disease Detection**
- **Fram Attachment Automation**
- **Crop Cycle Management**
- **AI based Spraying**

TRAx Controller

Terrain-Ready Autonomy Extension Controller





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R1 MINI

UNMANNED GROUND VEHICLE

OVERVIEW

Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is designed to simplify and accelerate the development of autonomous solutions — no robotics engineering degree required!
Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is

Traversability Analysis

Navigate with confidence using real-time terrain classification powered by RGB, LiDAR, and thermal sensors. ResQSense detects drivable surfaces, obstacles, and hazards—enabling safe maneuvering in unstructured and unpredictable terrain.

Vehicle Diagnostics

Ensure mission readiness with continuous health monitoring. From battery levels to sensor status, ResQSense provides real-time diagnostics and predictive alerts to minimize downtime and avoid critical failures.

Mission Command

Stay informed with a real-time, high-fidelity interface for visualising vehicle status, terrain mapping, and mission progress.The MCI delivers real-time mission progress through robust radio networks—keeping teams connected.

TECHNICAL SPECIFCATIONS

USB	4x USB 3.0	Operating Temperature	-20°C to +60°C	Infrared (IR) Cameras	Compatible via USB
Power Input	12V DC @ 8A	Vibration Resistance	Designed for environments	LiDAR	High-bandwidth supported
Protections	Overvoltage, reverse polarity,	Vehicle Type Support	Combustion engine-powered	RADAR	Kinematic Commands — velocity,angular
Enclosure	IP65-rated	RGB Camerras	Plug-and-play via USB	Operation Modes	Remote Control (RC), Autonomous
Wireless Communication	Dual-band RF (adaptive frequency	Function	Drive-by-wire & autonomy-enabling		

HUSKY AMP

THE FASTEST PATH TO BUILDING AUTONOMOUS ROBOTICS SOLUTIONS

Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is designed to simplify and accelerate the development of autonomous solutions — no robotics engineering degree required!

INTEGRATED SENSOR SUITE

Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is designed to

INCLUDED SENSORS

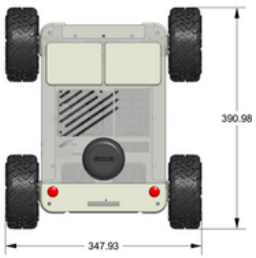
- Long Range WiFi and LTE
- Dual Antenna RTK GNSS
- 3D Directional LiDAR
- 2x Depth Cameras

EXAMPLES

- Long Range WiFi and LTE
- Dual Antenna RTK GNSS
- 3D Directional LiDAR
- 2x Depth Cameras

OTHER PACKAGES

Turnkey configuration packages offer everything needed to get started quickly. Additional sensors can be added or substituted upon request. Contact our team to learn about our custom integration services.



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CONTACT US

AutoMind Dynamics
1425 Strasburg Road, Unit 2A
Kitchener, Ontario N2R 1H2, Canada
TEL: 1-800-301-3863

TraX Controller

ARTPARK AI & Robotics Technology Park | AutoMind Dynamics

THE BRIDGE BETWEEN MANUAL AND AUTONOMOUS VEHICLES

The **TraX Controller** is a rugged, IP67-rated control box designed to convert conventional off-road/on-road vehicles into autonomy-ready drive-by-wire platforms. It integrates compute, control, and communication modules to seamlessly transition between remote and autonomous operation.

APPLICATIONS

- Remote Vehicle Operation & Inspection
- Drone Deployment & Charging Stations
- Night time operation
- Autonomous Off-Road / On-road Logistics
- Target detection and tracking
- AI based Surveillance/Reconnaissance

Dimensions	240x330x125mm	PORTS	Ethernet, 1x USB 3.0
Power Input	18V-24V DC @ 8A (Power from vehicle)	CAN Bus	2x CAN
Connector Type	Amphenol connectors	LiDAR	High-bandwidth supported (Ethernet/CAN)
Control I/O	CAN, UART, USB Ethernet	RGB / IR Cameras	Plug-and-play via USB
Operating Temperature	-20°C to +60°C	Enclosure	IP67-rated
Protections	Overvoltage, reverse polarity, current surge protection	Wireless Communication	Dual-band RF (3 km LOS tested)
RADAR	Configurable via CAN/serial interfaces	Serial	1x RS232, 1x UART

CAPABILITIES

Sensor Fusion

Compatible with LiDAR, RADAR, RGB, IR cameras

Peripheral Coordination

Supports robotic manipulators and drone docking/charging

Autonomous Mode

Velocity Control Integrated navigation, perception, planning

RC Mode

Human-in-the-loop via long-range RF

SYSTEM HIGHLIGHTS

Enables complete drive-by-wire control for throttle, brake, steering, and gear shift. Powered by Jetson for real-time AI perception and autonomy tasks. Dual-module architecture ensures reliable compute and precise actuation. Supports LiDAR, RADAR, RGB, and IR sensors for multi-sensor fusion. Operates seamlessly in both remote and autonomous modes.

CONTACT

Email -saadiyean@artpark.in

CONTROL BOX

THE FASTEST PATH TO BUILDING AUTONOMOUS ROBOTICS SOLUTIONS

Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is designed to

EXAMPLES

- Long Range WiFi and LTE
- Dual Antenna RTK GNSS
- 3D Directional LiDAR
- 2x Depth Cameras

WHY TRAX

- Unified compute–control arc
- AI-ready with ROS2 and ONN support
- Modular integration for any v type
- Field-proven ruggedness

DIMENSIONS	390 x 348 x 188 mm (15.35 x 13.7 x 7.4 in)	RUN TIME	Heavy usage: 2 hrs Basic Usage: 4 hrs	COMMUNICATION	Ethernet, USB 3.0
WEIGHT	8 kg (17.6 lbs)	DRIVE POWER	80 W	DRIVERS AND APIs	Packaged with ROS2 Humble
MAX. PAYLOAD	5 kg (11 lbs)	BATTERY	12 Ah (6Ah X 2 NOS) Swap-able Battery	CONTROL MODES	Kinematic Commands — velocity,angular velocity Open Loop
MAX SPEED	1.36 m/s (4.9 Km/h)	CHARGE TIME	1 hour per battery	FEEDBACK	Battery, motor current Wheel velocity and IMU
CLEARENCE	88.5 mm (3.5 in)	USER POWER	5 V at 5 A, 12 V at 5 A	SUPPORTED OPERATING SYSTEMS	Ubuntu 22.04LTS (JetPack 6.2)

GROUND ROBOTICS TUTORIAL

POINT TO POINT NAVIGATION

SLAM

ADAS

SENSOR FUSION

OBJECT DETECTION

OBSTACLE AVOIDANCE

CONTACT

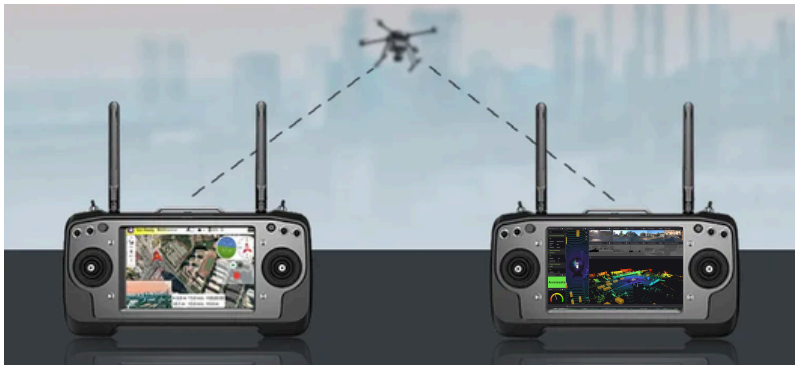
Ph. NO - 70006275705

Email -qiranuls@iisc.ac.in

TECHNICAL SPECIFCATIONS

TRAx Controller™

Terrain-Ready Autonomy extension Controller



USB	4x USB 3.0
Power Input	12V DC @ 8A
Protections	Overvoltage, reverse polarity, current surge protection
Enclosure	IP65-rated
Wireless Communication	Dual-band RF (adaptive frequency hopping), 3 km LOS tested

Operating Temperature	-20°C to +60°C
Vibration Resistance	Designed for off-road environments
Vehicle Type Support	Combustion engine-powered off-road/ on-road platforms
RGB Camerras	Plug-and-play via USB
Function	Drive-by-wire & autonomy-enabling control unit for off-road vehicle

Infrared (IR) Cameras	Compatible via USB
LiDAR	High-bandwidth supported (Ethernet/CAN)
RADAR	Kinematic Commands — velocity,angular velocity Open Loop
Operation Modes	Remote Control (RC), Autonomous

APPLICATIONS

Remote Vehicle Operation & Inspection

Target detection and tracking (ICB v2)

AI based Surveillance/Reconnaissance (ICV v2)

Night time operation

Autonomous Off-Road Logistics (Upcoming)

CONTACT

Ph. NO - 70006275705

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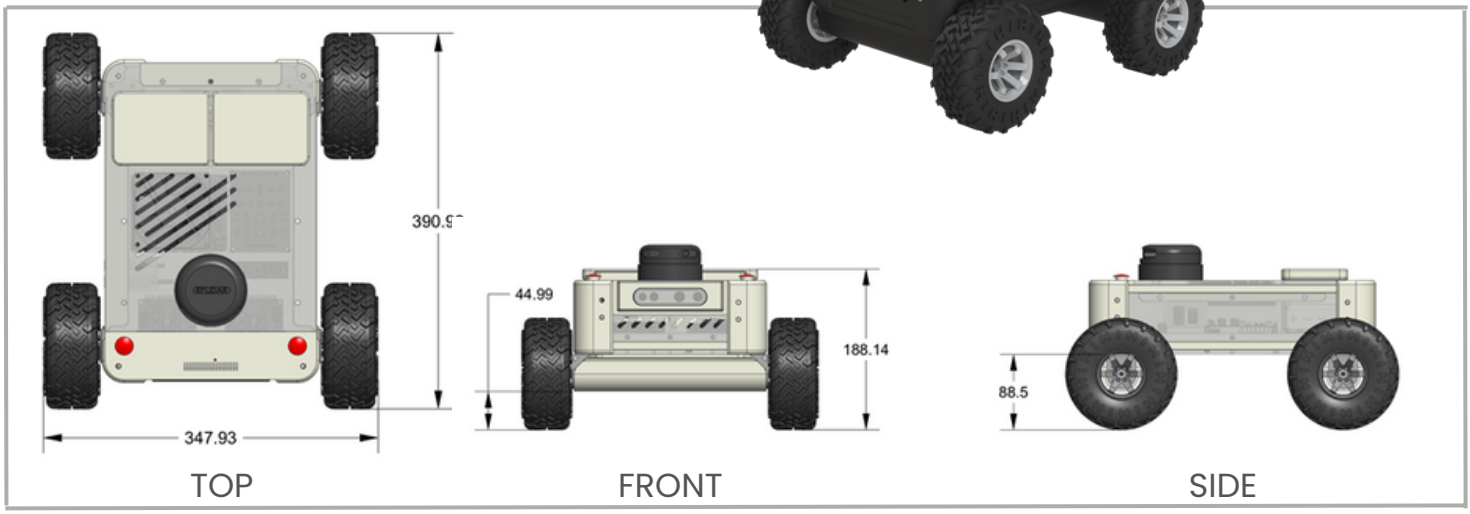
TECHNICAL SPECIFCATIONS

R1 MINI™

OUTDOOR VERSION

AUTONOMOUS GROUND VECHILE



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GROUND ROBOTICS TUTORIAL

POINT TO POINT NAVIGATION	SLAM	ADAS	SENSOR FUSION
OBJECT DETECTION	OBSTACLE AVOIDANCE		

CONTACT

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R1 MINI

OUTDOOR VERSION

AUTONOMOUS GROUND VECHILE

THE BRIDGE BETWEEN MANUAL AND AUTONOMOUS VEHICLES

The **TraX Controller** is a rugged, IP65-rated control box designed to convert conventional off-road vehicles into autonomy-ready drive-by-wire platforms. It integrates compute, control, and communication modules to seamlessly transition between remote and autonomous operation.

APPLICATIONS

- Remote Vehicle Operation & Inspection
- Drone Deployment & Charging Stations
- Night time operation
- Autonomous Off-Road Logistics (Upcoming)
- Target detection and tracking (ICB v2)
- AI based Surveillance/Reconnaissance (ICV v2)

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GROUND ROBOTICS TUTORIAL

POINT TO POINT NAVIGATION

SLAM

ADAS

SENSOR FUSION

OBJECT DETECTION

OBSTACLE AVOIDANCE

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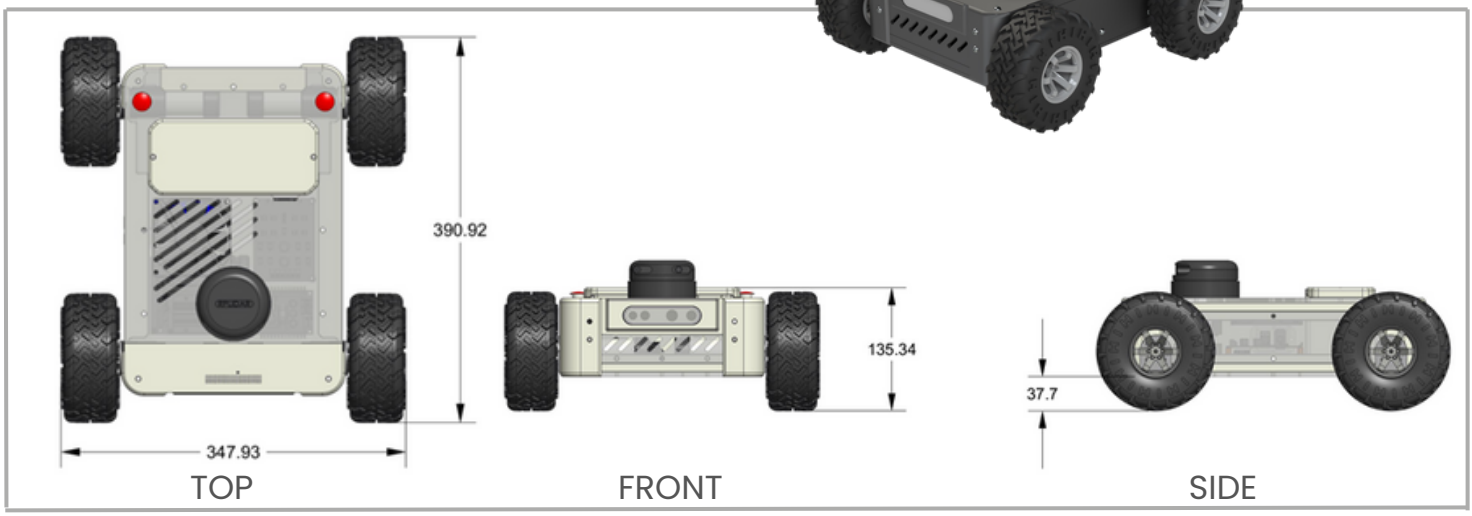
TECHNICAL SPECIFCATIONS

R1 MINI™

INDOOR VERSION

AUTONOMOUS GROUND VECHILE



DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)	RUN TIME	Heavy usage: 3 hrs Basic Usage: 5 hrs	COMMUNICATION	Ethernet, USB 3.0
WEIGHT	8 kg (17.6 lbs)	DRIVE POWER	80 W	DRIVERS AND APIs	Packaged with ROS2 Humble
MAX. PAYLOAD	5 kg (11 lbs)	BATTERY	15 Ah Lithium Ferrous Phosphate	CONTROL MODES	Kinematic Commands — velocity,angular velocity Open Loop
MAX SPEED	1.36 m/s (4.9 Km/h)	CHARGE TIME	1.5 hour	FEEDBACK	Battery, motor current Wheel velocity and IMU
CLEARENCE	37.7 mm (1.5 in)	USER POWER	5 V at 5 A, 12 V at 5 A	SUPPORTED OPERATING SYSTEMS	Ubuntu 22.04LTS (JetPack 6.2)

GROUND ROBOTICS TUTORIAL

POINT TO POINT NAVIGATION | **SLAM** | **ADAS** | **SENSOR FUSION**

OBJECT DETECTION | **OBSTACLE AVOIDANCE**

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TECHNICAL SPECIFCATIONS

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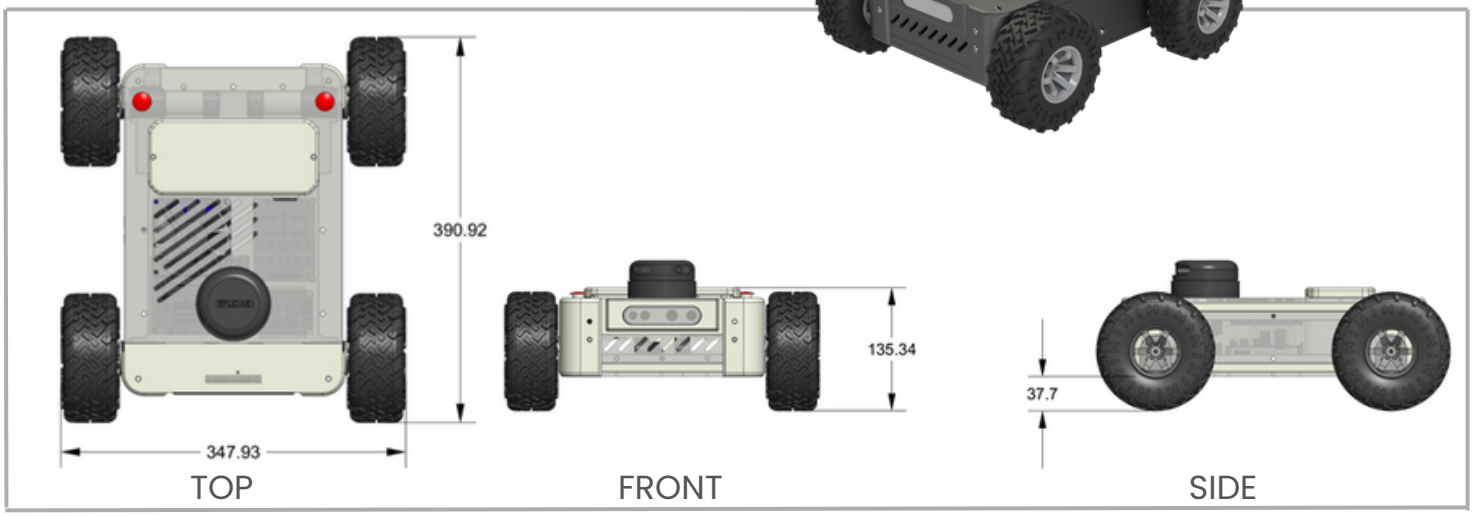
INDOOR VERSION

AUTONOMOUS GROUND VECHILE



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OBJECT DETECTION

OBSTACLE AVOIDANCE

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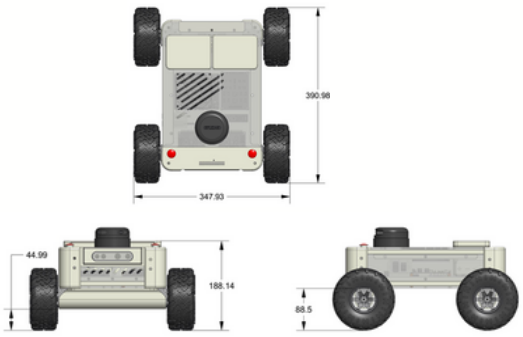
MINI ROVER I

UNMANNED
GROUND VECHILE

OUTDOOR VERSION

Sebastian Bennet



DIMENSIONS	390 x 348 x 188 mm (15.35 x 13.7 x 7.4 in)
WIEGHT	8 kg (17.6 lbs)
GROUND CLEARENCE	88.5 mm (3.5 in)
TIRE DIAMETER	4 x 130 mm (5 in) OFFROAD TIRES
MAX. PAYLOAD	5 kg (11 lbs)
ALL-TERRAIN PAYLOAD	2 kg (4.4 lbs)
MAX. SPEED	1.36 m/s (4.9 Km/h)
MOTORS	4 x 20 W DC MOTORS
DRIVE POWER	80 W

BATTERY CHEMISTRY	Lithium Ferrous Phosphate
CAPACITY	12 Ah (6Ah X 2 NOS) - Swap-able Battery
CHARGE TIME	1 hour per battery
RUN TIME	Heavy usage: 2 hrs Basic Usage: 4 hrs
USER POWER	5 V at 5 A, 12 V at 5 A
CONTROL MODES	Kinematic Commands — velocity, angular velocity Open Loop
FEEDBACK	Battery, motor current Wheel velocity and IMU
COMMUNICATION	Ethernet, USB 3.0
DRIVERS AND APIs	Packaged with ROS2 Humble
SUPPORTED OPERATING SYSTEMS	Ubuntu 22.04LTS (JetPack 6.2)

CONTACT US:

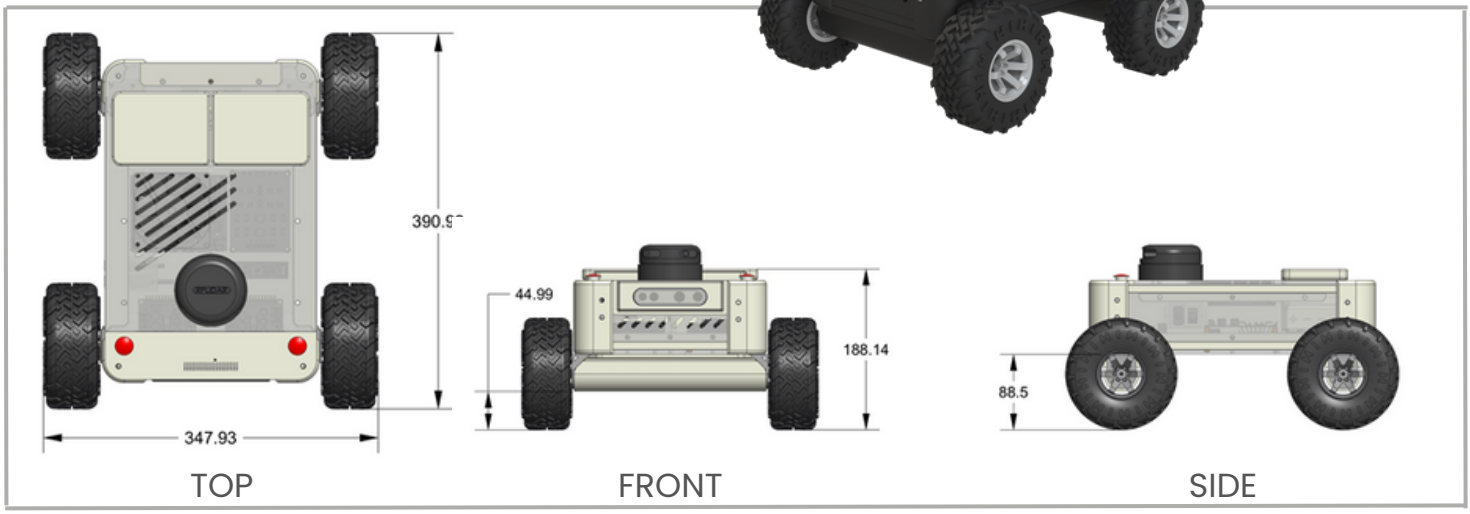
Brocelle Company

TECHNICAL SPECIFCATIONS

R1 MINI™

OUTDOOR VERSION

AUTONOMOUS GROUND VECHILE



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GROUND ROBOTICS TUTORIAL

POINT TO POINT NAVIGATION	SLAM	ADAS	SENSOR FUSION
OBJECT DETECTION	OBSTACLE AVOIDANCE		

CONTACT

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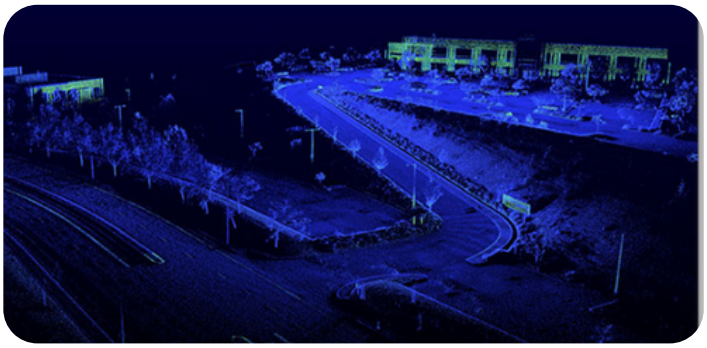
Email -qiranuls@iisc.ac.in

ResQSense Suite

Intelligent Mobility • Active Awareness • Total Confidence

ABOUT

ResQSense is a rugged, all-in-one sensor and perception suite designed to enhance safety and situational awareness in disaster zones. Built for off-road, urban, and indoor environments, it empowers rescue vehicles to operate with confidence in the most challenging condition in both day and night operation.



FEATURES

Traversability Analysis Suite

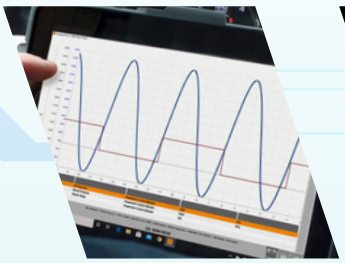
Navigate with confidence using real-time terrain classification powered by RGB, LiDAR, and thermal sensors. ResQSense detects drivable surfaces, obstacles, and hazards—enabling safe maneuvering in unstructured and unpredictable terrain.

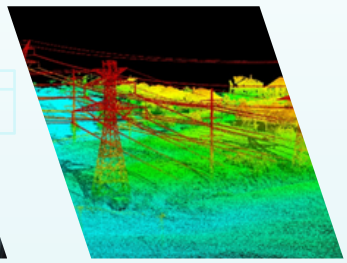
Vehicle Diagnostics

Ensure mission readiness with continuous health monitoring. From battery levels to sensor status, ResQSense provides real-time diagnostics and predictive alerts to minimize downtime and avoid critical failures.

Mission Command Interface

Stay informed with a real-time, high-fidelity interface for visualising vehicle status, terrain mapping, and mission progress.The MCI delivers real-time mission progress through robust radio networks—keeping teams connected.





STANDARD VERSION

- Onboard Edge Processing
- DAY & NIGHT OPERATION
- RGB + Thermal Vision
- Terrain-Aware Sensing
- Advanced Multi-Sensor Fusion
- Dynamic Terrain Reconstruction
- Data Logging and Playback
- LiDAR
- Mission Command Interface

ResQSense Suite

Intelligent Mobility • Active Awareness • Total Confidence

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ResQSense is a rugged, all-in-one sensor and perception suite designed to enhance safety and situational awareness in disaster zones. Built for off-road, urban, and indoor environments, it empowers rescue vehicles to operate with confidence in the most challenging condition in both day and night operation.

SPECIFICATIONS

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STAN

Onboard Edge Processing

DAY & NIGHT OPERATION

RGB + Thermal Vision

Terrain-Aware Sensing

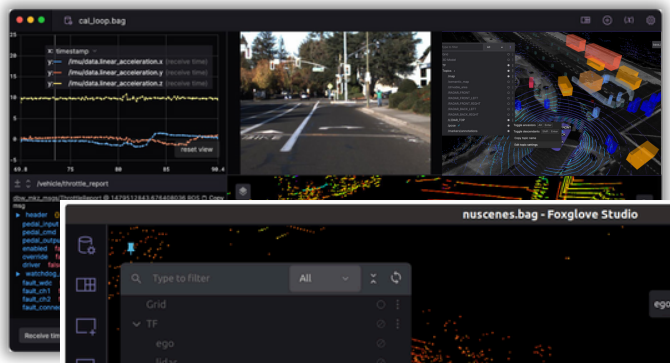
Advanced Multi-Sensor Fusion

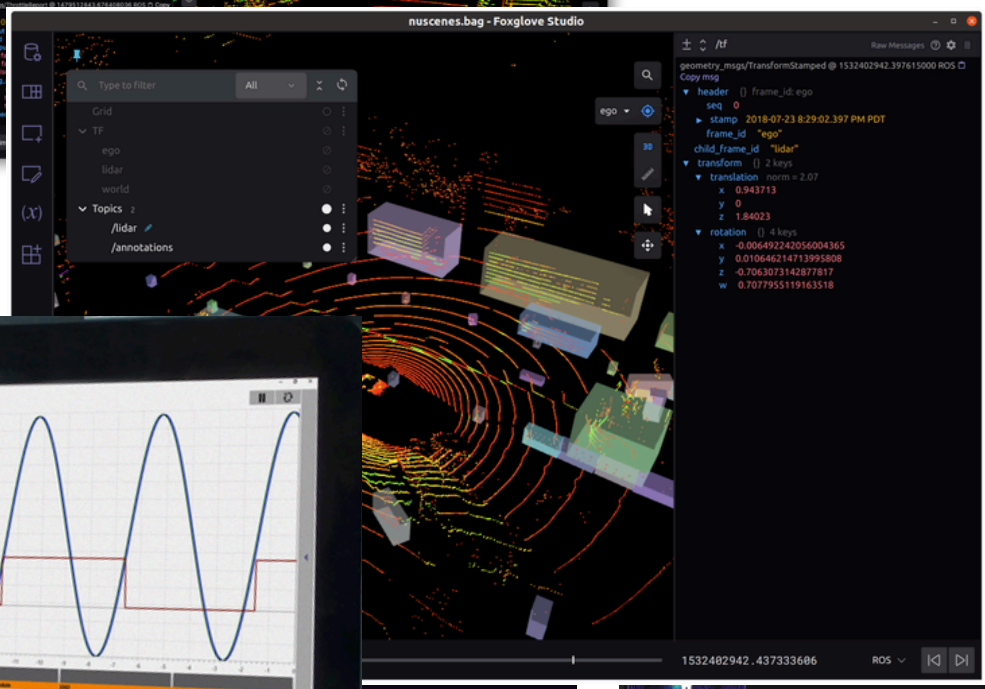
Dynamic Terrain Reconstruction

Data Logging and Playback

LiDAR

Mission Command Interface

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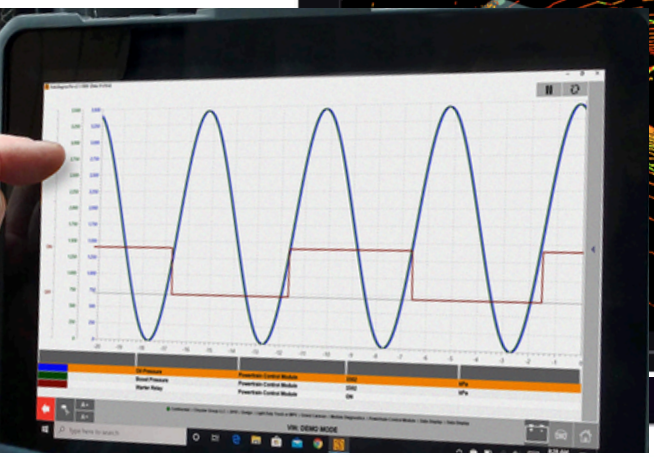

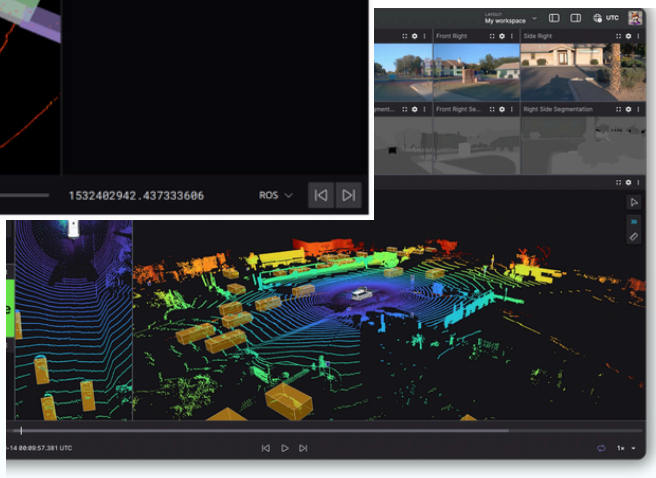

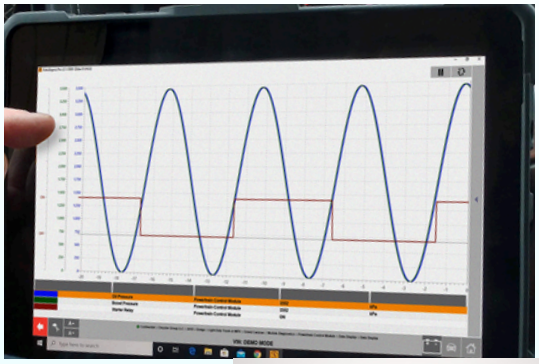

TRAX

R1 MINI

OUTDOOR VERSION

AUTONOMOUS GROUND VECHILE

ARTPARK
AI & Robotics Technology Park

भारतीय विज्ञान संस्थान

SYSTEM HIGHLIGHTS

Built on the rugged and customizable Husky A300 mobile base, Husky AMP (Autonomous Mobile Platform) is designed to simplify and accelerate the development of autonomous solutions

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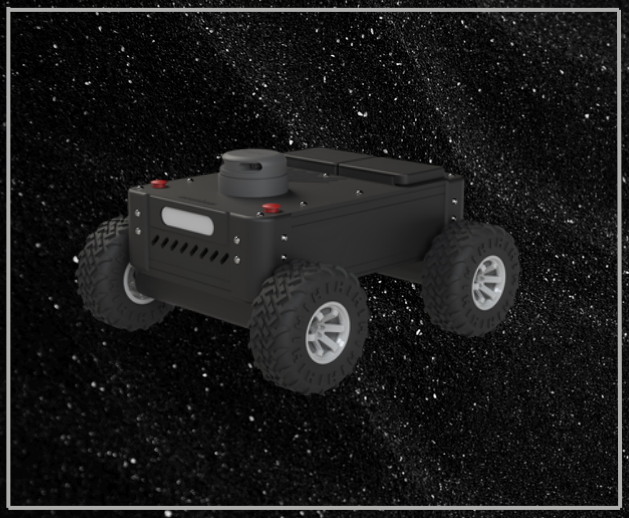

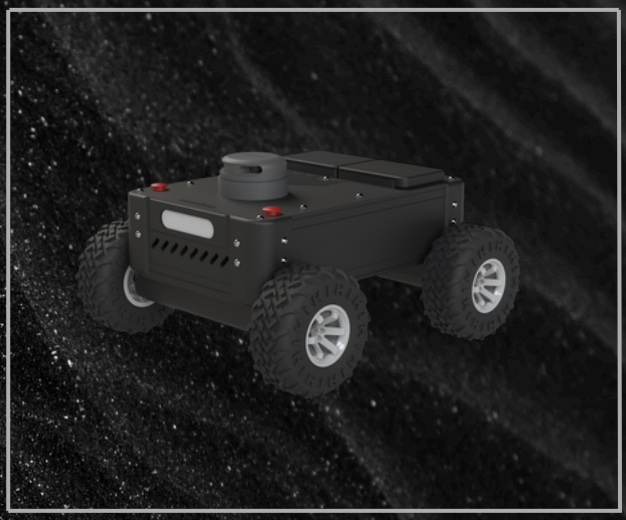

DIMENSIONS	390 x 348 x 188 mm (15.35 x 13.7 x 7.4 in)	RUN TIME	Heavy usage: 2 hrs Basic Usage: 4 hrs	COMMUNICATION	Ethernet, USB 3.0
WEIGHT	8 kg (17.6 lbs)	DRIVE POWER	80 W	DRIVERS AND APIs	Packaged with ROS2 Humble
MAX. PAYLOAD	5 kg (11 lbs)	BATTERY	12 Ah (6Ah X 2 NOS) Swap-able Battery	CONTROL MODES	Kinematic Commands — velocity,angular velocity Open Loop
MAX SPEED	1.36 m/s (4.9 Km/h)	CHARGE TIME	1 hour per battery	FEEDBACK	Battery, motor current Wheel velocity and IMU

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OUTDOOR VERSION

AUTONOMOUS GROUND VECHILE

ABOUT US

ResQSense is a rugged, all-in-one sensor and perception suite designed to enhance safety and situational awareness in disaster zones. Built for off-road, urban, and indoor environments, it empowers rescue vehicles to operate with confidence in the most challenging condition in both day and night operation.

OUR PRODUCTS

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ALGO

- Remote Vehicle Operation & Inspection
- Drone Deployment & Charging Stations
- Night time operation
- Autonomous Off-Road Logistics (Upcoming)
- Target detection and tracking (ICB v2)
- AI based Surveillance/Reconnaissance (ICV v2)

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Sl. No	PARAMETER	RATING
1	DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)
2	DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)
3	DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)
4	DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)
5	DIMENSIONS	390 x 348 x 135 mm (15.35 x 13.7 x 5.3 in)

R1 MINI™

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SUPPORTED BY

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MINISTRY OF SCIENCE AND TECHNOLOGY
सत्यमेव जयते

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MINISTRY OF SCIENCE AND TECHNOLOGY
सत्यमेव जयते

DST@50 1971-2021 DEPARTMENT OF SCIENCE & TECHNOLOGY

INCUBATED AT

ARTPARK AI & Robotics Technology Park

AIRL Department of Aerospace Engineering Indian Institute of Science, Bangalore

INDIAN INSTITUTE OF SCIENCE
भारतीय विज्ञान संस्थान

CONTACT

Ph. NO - 70006275705

Email -qiranuls@iisc.ac.in

Email -qiranuls@iisc.ac.in

Email -qiranuls@iisc.ac.in

Email -qiranuls@iisc.ac.in

Email -qiranuls@iisc.ac.in